Integrative In Silico Workflow for Prioritization of Enzyme Drug Targets in *Mycobacterium tuberculosis*

A subtractive proteomics and in silico characterization study

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BACKGROUND

Antibiotic Resistance: A Growing Global Threat

- Increasing emergence of multidrug-resistant (MDR) pathogens
- Major challenge in effective treatment of infectious diseases

WHO 2024 Priority Pathogen List

- Updated list to guide R&D of new antibiotics
- Classified into **Critical, High, and Medium priority groups**

Critical Priority Pathogens

- Includes carbapenem-resistant *Acinetobacter baumannii* & *Pseudomonas aeruginosa*
- New addition: **Rifampicin-resistant *Mycobacterium tuberculosis***

Why Focus on Rifampicin-resistant *M. tuberculosis*?

- Global TB burden with rising resistance
- Limited treatment options and urgent need for novel therapeutics

Critical group



Enterobacterales
carbapenem-resistant



Enterobacterales
third-generation cephalosporin-resistant



Acinetobacter baumannii
carbapenem-resistant



Mycobacterium tuberculosis, rifampicin-resistant[a]

[a]RR-TB was included after an independent analysis with parallel criteria and subsequent application of an adapted MCDA matrix.

High group



Salmonella Typhi
fluoroquinolone-resistant



Shigella spp.
fluoroquinolone-resistant



Enterococcus faecium
vancomycin-resistant



Pseudomonas aeruginosa
carbapenem-resistant



Non-typhoidal *Salmonella*
fluoroquinolone-resistant



Neisseria gonorrhoeae
third-generation cephalosporin, and/or fluoroquinolone-resistant



Staphylococcus aureus
methicillin-resistant

Medium group



Group A Streptococci
macrolide-resistant



Streptococcus pneumoniae
macrolide-resistant



Haemophilus influenzae
ampicillin-resistant



Group B Streptococci
penicillin-resistant

Mycobacterium tuberculosis: A Critical Threat

I. About the Pathogen

- Causative agent of **tuberculosis (TB)**, one of the top 10 global killers
- Infects primarily the lungs, but also extrapulmonary sites
- Highly contagious, transmitted via **airborne droplets.**

II. Why is it a Critical Threat?

- WHO reports **>10 million new TB cases annually**
- Emergence of **MDR-TB** and **XDR-TB** strains
- **Rifampicin-resistant TB (RR-TB)** classified as *critical priority pathogen* (WHO 2024).

III. Current Treatment Options

- First-line: Rifampicin, Isoniazid, Ethambutol, Pyrazinamide
- Second-line: Fluoroquinolones, Aminoglycosides, Linezolid

IV. Limitations of Existing Therapies

- Long treatment duration (6–24 months)
- High toxicity and side effects
- Increasing treatment failures due to **drug resistance**
- No highly effective **preventive vaccine** beyond BCG (limited efficacy in adults)

Our Integrative Pipeline for target identification

Why our Pipeline Helps?

I. Systematic Filtering → High-Confidence Target

- Starts from **all genomes → core proteome → non-homologous → essential → virulent → membrane → enzymes → hub proteins → structural/chemical filters**.
- Ensures **step-by-step narrowing** → from **34,765 proteins → 1 final druggable target**.
- Reduces false positives and avoids chasing redundant/non-essential proteins.

II. Integrates Multiple Computational Domains

- Combines **comparative genomics, subtractive proteomics, network biology, structural bioinformatics** in one workflow.
- Cross-validates results across tools (BLAST, Prodigal, STRING, CELLO/PSORT, etc.).
- Moves logically from **sequence → structure → function → druggability**.

III. Druggability & Clinical Relevance

Focus on **essential, virulent, non-host homologous, membrane enzymes** → ensures the target is:

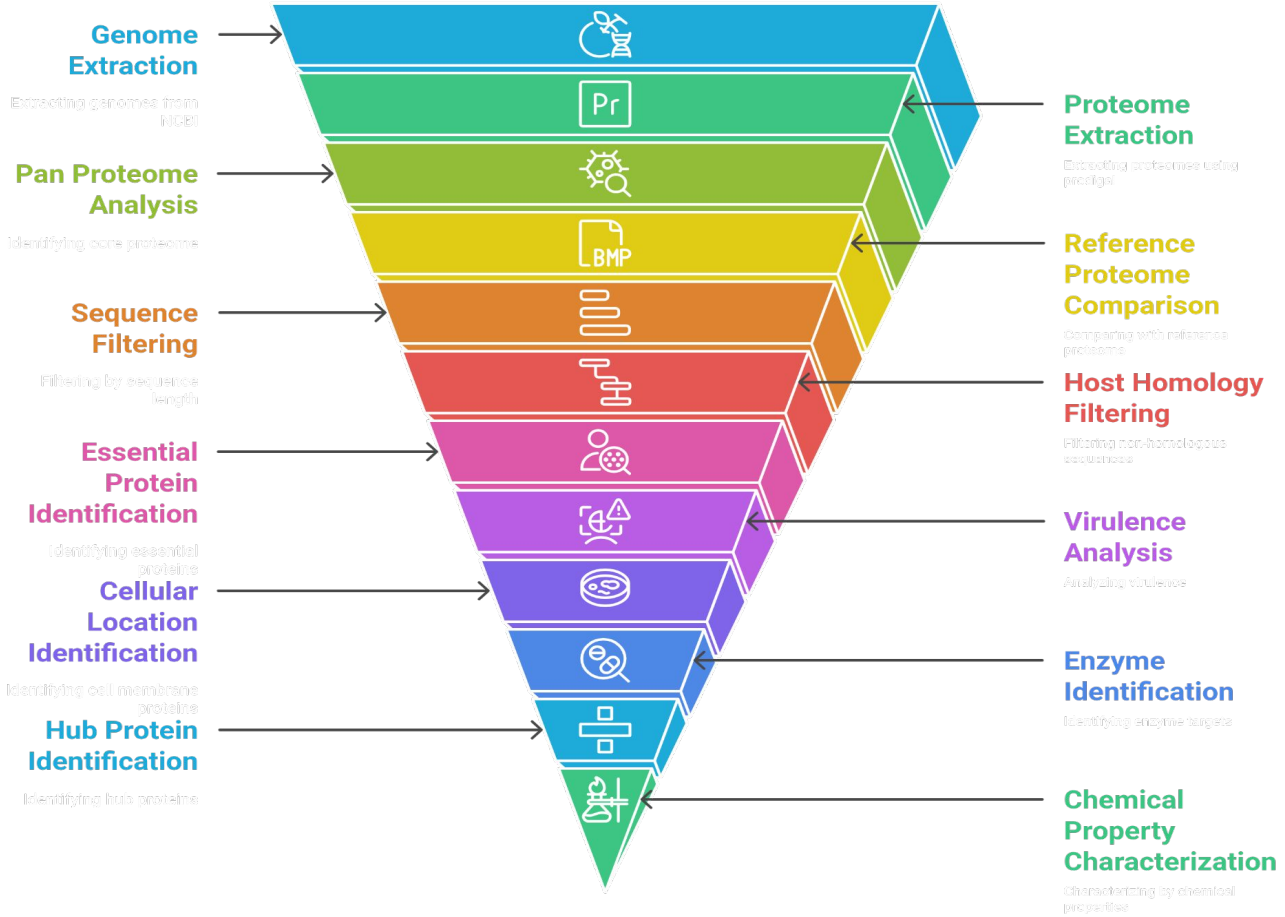
- **Unique to pathogen** (no host homology → reduced side effects).
- **Accessible** (membrane localization → drug accessibility).
- **Critical for survival** (essential + virulence → higher therapeutic relevance).

Leads to **high translational value** for downstream **drug/vaccine design**.

IV. End-to-End Pipeline

Doesn't stop at target discovery → continues to **structure prediction (2D/3D)**, **active site analysis**, and **docking readiness**.

Makes it a **complete drug discovery framework**, not just a theoretical pipeline.


Genome Extraction
Extracting genomes from NCBI
Pan Proteome Analysis
Identifying core proteome
Sequence Filtering
Filtering by sequence length
Essential Protein Identification
Identifying essential proteins
Cellular Location Identification
Identifying cell membrane proteins
Hub Protein Identification
Identifying hub proteins
Pr
BMP
Proteome Extraction
Extracting proteomes using prodigal
Reference Proteome Comparison
Comparing with reference proteome
Host Homology Filtering
Filtering non-homologous sequences
Virulence Analysis
Analyzing Virulence
Enzyme Identification
Identifying enzyme targets
Chemical Property Characterization
Characterizing by chemical properties

Step 1: Pan Proteome Analysis

Defining the core proteome of *Mycobacterium tuberculosis* from a representative sample of genomes

Genome collection

- Total *complete* genomes available in NCBI: **1044**.
- From these, we **randomly selected 10 genomes** for downstream pan-genome analysis.
- (⚠️ These were not filtered using CheckM — we directly used them as they were available in NCBI.)

Proteome prediction

- For each genome, **Prodigal** was used to predict coding sequences (CDS) and generate corresponding **protein FASTA (`.faa`) files**.
- Each proteome contained **~4000 proteins**.

All-vs-all BLASTp

- Built a BLAST database for each proteome (`makeblastdb`).
- Ran all-vs-all protein comparisons: **10 × 9 = 90 BLASTp runs**.
- Scale of computation:
    - ~4000 proteins/query × 4000 proteins/subject ≈ **16 million alignments per run**.
    - Across 90 runs ≈ **1.4 billion alignments** total.
- Used `-num_threads 4` for parallelization within runs.
- This step was the computational backbone for clustering.

Clustering into orthologous groups

- Filtered BLASTp hits with **stringent thresholds**:
    - **Identity ≥ 90%**
    - **Coverage ≥ 80%**
    - **E-value ≤ 1e-5**
- Built clusters (protein families) using a greedy single-linkage approach.
- Generated:
    - **core clusters**: present in all 10 genomes.
    - **accessory clusters**: present in some but not all genomes.
    - **singletons**: unique to one genome.

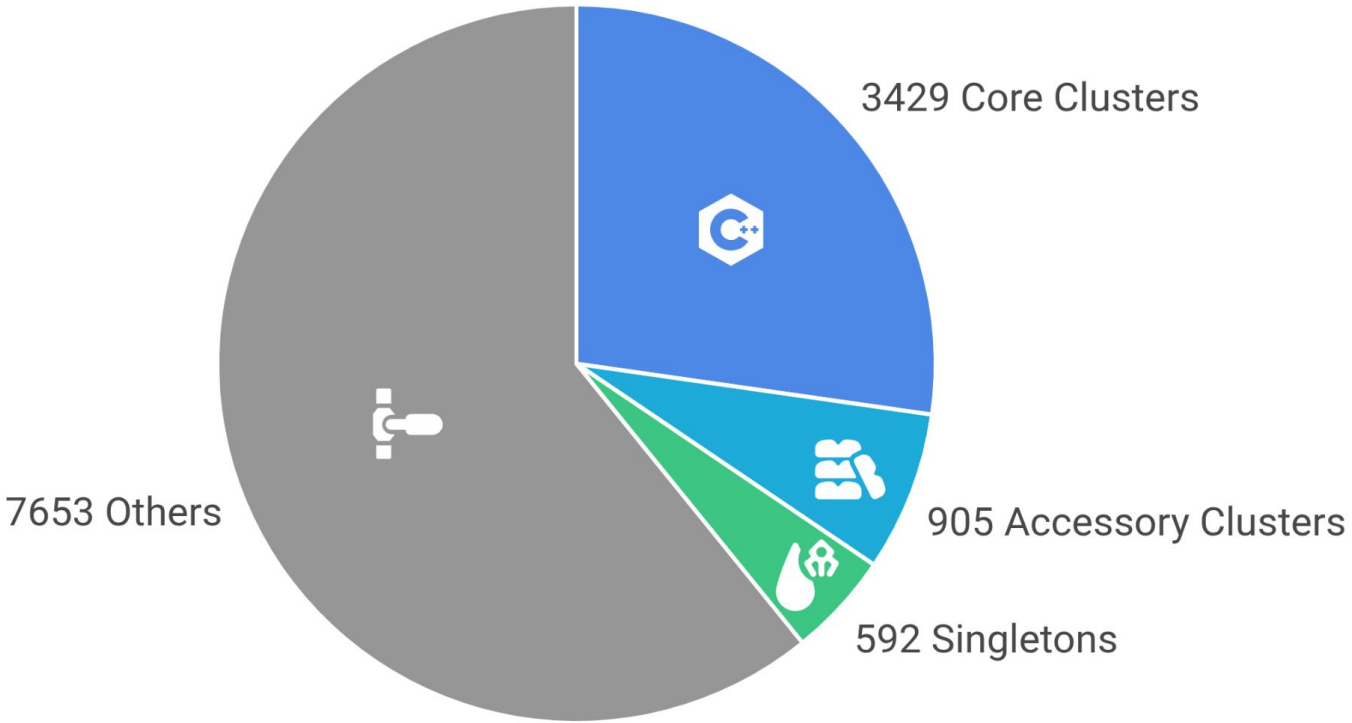
Results of clustering

- **Total clusters**: 12,579
- **Core clusters**: 3,429
- **Accessory clusters**: 905
- **Singletons**: 592

Core proteome FASTA output

- `output_faa/core.faa` contained **34,765 protein sequences**.
- ⚠️ Important clarification:
    - This is **not 3,429 unique proteins**.
    - Instead, this file includes **all individual protein sequences from all 10 genomes** that belong to the 3,429 “core clusters.”
    - Example: if a core cluster has one protein copy per genome, that cluster contributes 10 protein sequences → hence ~34k total.

Distribution of Protein Family Clusters (clusters)



KEY INSIGHTS :

The **real “core orthologs”** = **3,429 clusters** (i.e., one representative sequence per cluster).

The **core FASTA (34,765 sequences)** = all instances of those clusters across the 10 genomes.

We proceeded downstream with the **unique representatives** (3,429) to avoid redundancy.

Problem with Step -1

- The **10 genomes used for pan-genome analysis** were chosen **randomly** from the 1044 available in NCBI.
- ⚠️ These were **not pre-filtered by CheckM** (quality control metric for completeness/contamination).
- As a result, the “core proteome” obtained might include **false positives/negatives** due to sequencing/assembly errors in low-quality genomes.

Step 2: Core Proteome v/s Reference Proteome

Reference Proteome Validation of Core Proteome

Solution : To ensure high reliability in downstream analysis:

1. **Selected a high-quality reference genome**
   - Used the **reference genome of *Mycobacterium tuberculosis***.
   - CheckM quality: **97%** (high completeness, low contamination).
   - This guarantees a trusted baseline proteome.

2. **BLASTp: Reference proteome vs Core proteome**
   - Query: Reference proteome (≈4000 proteins).
   - Subject: Core proteome (34,765 sequences from 3,429 clusters).
   - Filtering criteria for “best hit, high confidence”:
     - **Identity ≥ 90%**
     - **Query coverage ≥ 80%**
     - **E-value ≤ 1e-05**

3. **Why this comparison?**
   - To extract **only those proteins that are both in the high-quality reference genome and also present across all 10 genomes (core set)**.
   - This step **removes redundancy** and ensures we carry forward only **true core-confirmed proteins**.

Results

- **Total distinct query sequences in BLAST results:** 3,613
- **Unique core sequences matched:** 3,393
- **Best-hit HIGH-MATCH count (stringent filters): 429**

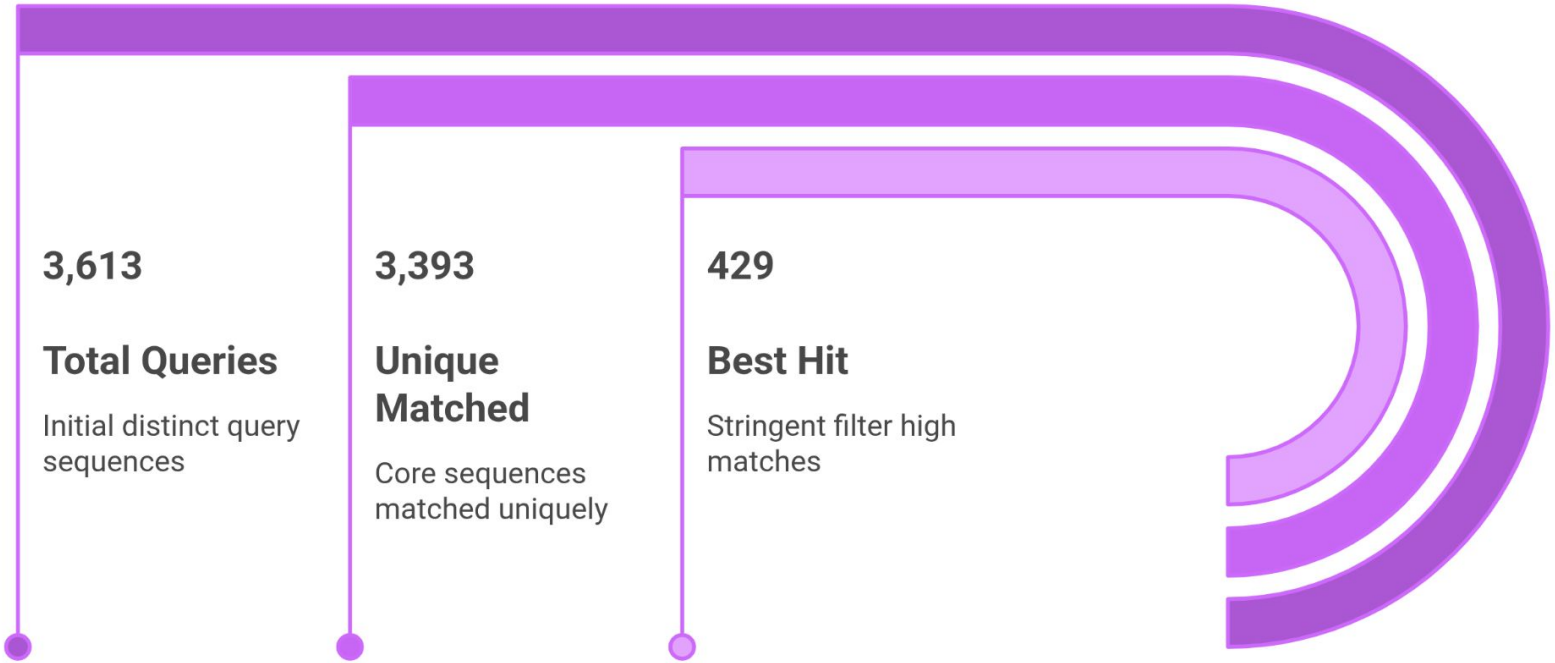
👉 These **429 proteins** are our **core-confirmed, high-confidence proteins**, forming the foundation for all downstream analysis.

Key Insight

- Instead of relying on the raw “core” from random genomes (possibly noisy), we anchored our analysis to a **high-quality reference genome**.

- This ensures that our pipeline is **robust against low-quality assemblies** and the downstream targets are **biologically reliable**.

BLAST Query Sequence Analysis



Step 3: Length Based Filtering

Length Based Filtering of Core Proteins

Why did we do this?

- Very **short proteins (<100 amino acids)** are often:
  - **Hypothetical proteins** with no clear function.
  - **Fragments** of mis-annotated or incomplete genes.
  - Not reliable drug targets because they may lack stable structural domains.
- Since we aim to identify **robust drug targets**, we excluded sequences shorter than 100 aa.

Process

1. Input: **429 high-confidence core-confirmed proteins** (from Step 2).
2. Wrote a Python script (`filter_by_length.py`) to:
   - Parse each sequence.
   - Measure its amino acid length.
   - Keep only proteins ≥100 aa.
   - Discard shorter ones.

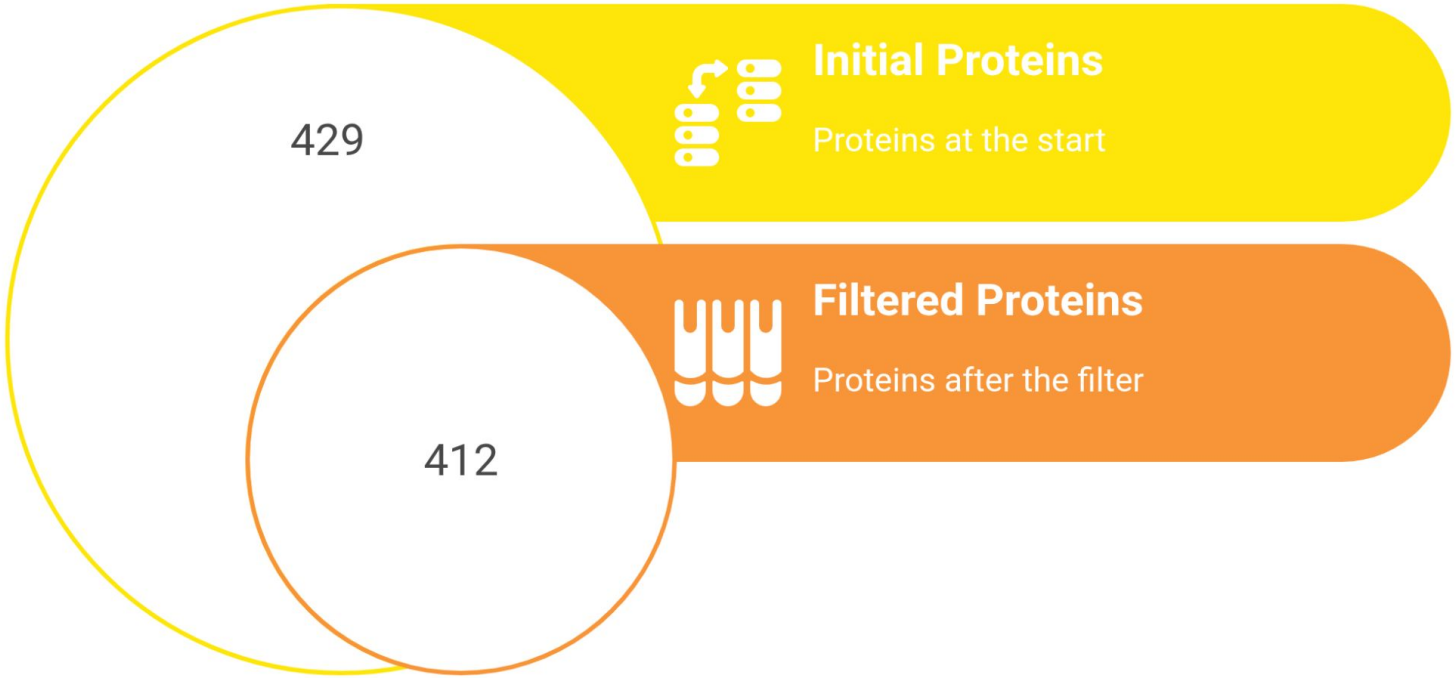
Results

- **Proteins kept:** 412
- **Proteins removed (<100 aa):** 17
- **Output file:** `core_filtered.faa`

Key Insight

- This step ensured that only **biologically meaningful, structurally viable proteins** were retained for downstream filtering.
- It helped us focus our computational and experimental resources on targets **more likely to be functionally relevant drug candidates**.

Protein Filtering by Amino Acid Count



Step 4: Host Homology

Host Homology

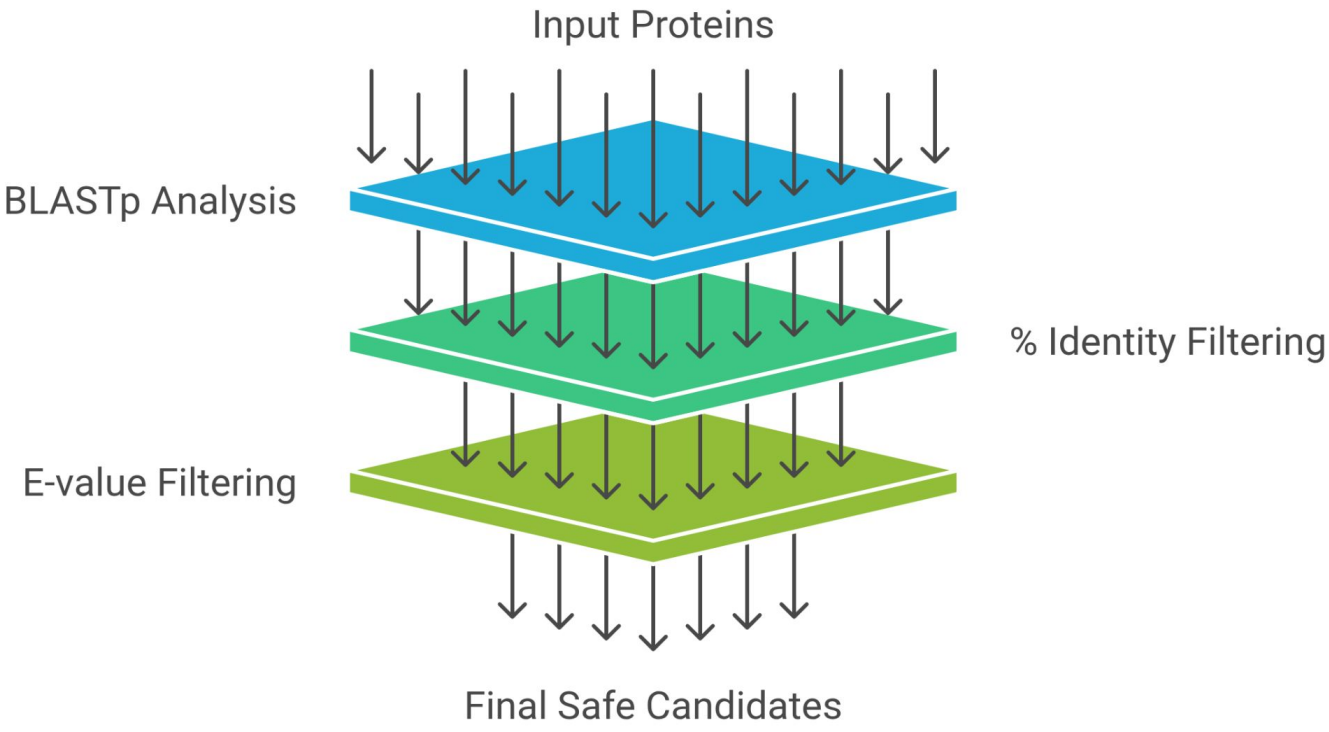
Why we did it (Relevance):

- One of the crucial steps in reverse vaccinology is to ensure that the predicted proteins/epitopes are **not homologous to the host (human)**.
- This prevents the risk of **autoimmunity** and **cross-reactivity** with human proteins.
- Ensures **safety** and improves **specificity** of potential vaccine candidates.

How we performed it (Process):

1. The shortlisted proteins from the previous step (non-paralogous & essential proteins) were taken as input.
2. These protein sequences were subjected to **BLASTp against the human proteome** (NCBI RefSeq / Uniprot database).
3. Parameters were set to identify significant similarity (e.g., E-value ≤ 1e-4, % identity ≥ 35–40% considered significant).
4. 369 Proteins showing **no significant homology** with human proteins were retained.

Protein Candidate Selection Process



Step 5: Essential Protein Identification

Essential Protein Analysis

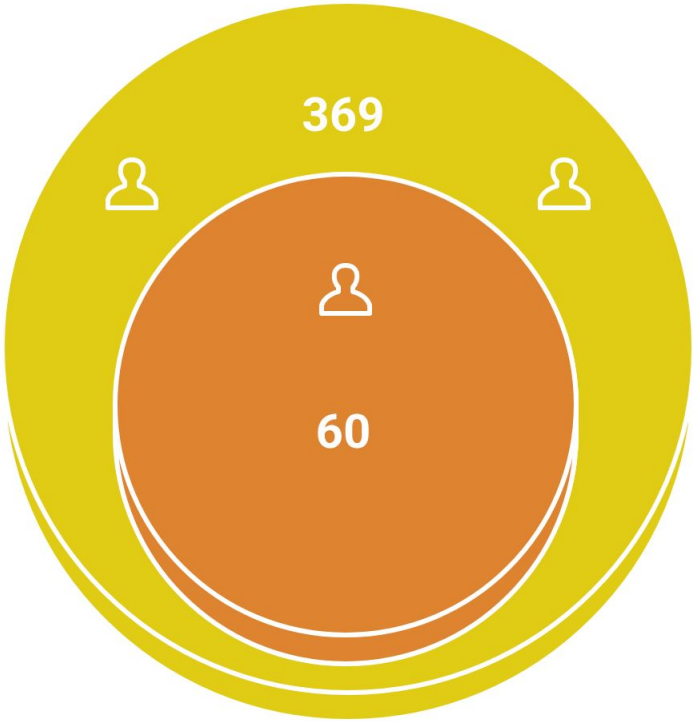
Why we did it (Relevance):

- Essential proteins are crucial for the **survival and growth** of the pathogen.
- Targeting such proteins increases the likelihood that the pathogen will not be able to develop compensatory mechanisms.
- Helps us filter out proteins that are most promising for **vaccine development or drug targets**.

How we performed it (Process):

1. The **369 core, non-human homolog proteins** (from Step 4) were taken as input.
2. These were screened against the **Database of Essential Genes (DEG)**.
3. BLASTp search was used with optimized parameters (e.g., E-value ≤ 1e-10, bit score > 100).
4. Proteins that showed significant matches with essential gene entries were retained.

Protein Analysis Workflow



Total Proteins

Initial protein count

Essential Proteins

Proteins after BLASTp search

Step 6: Virulent Protein Identification

Virulence Analysis

Why we did it (Relevance):

- Virulence factors are proteins that help the pathogen **invade, colonize, and evade the host immune system**.
- Identifying virulent proteins among essential ones helps us prioritize proteins that are not only required for survival but also play a **direct role in pathogenicity**.
- Such proteins make **ideal targets** for therapeutic interventions or vaccine design.

How we performed it (Process):

1. Problem: **VFDB / pBIT online** was not accessible and **VirulentPred2.0 standalone tools** requiring Linux system (not available). Online VirulentPred2.0 tool had a **sequence input limit (≤10 sequences)**, unsuitable for our dataset.

2. To overcome this, we used a **custom database approach**:

    - Previously, we had identified **virulent proteins in *M. tuberculosis*** using pBIT (when it was accessible).
    - These virulent protein sequences were compiled into a **local BLAST database**.

3. The **60 essential proteins** (from Step 5) were queried against this database using **BLASTp**.

4. All 60 Proteins with significant similarity to known virulent proteins were retained.



60 Virulent Proteins

Step 7: Identification of Cellular Location

Subcellular Location Filtering

Why we did it (Relevance):

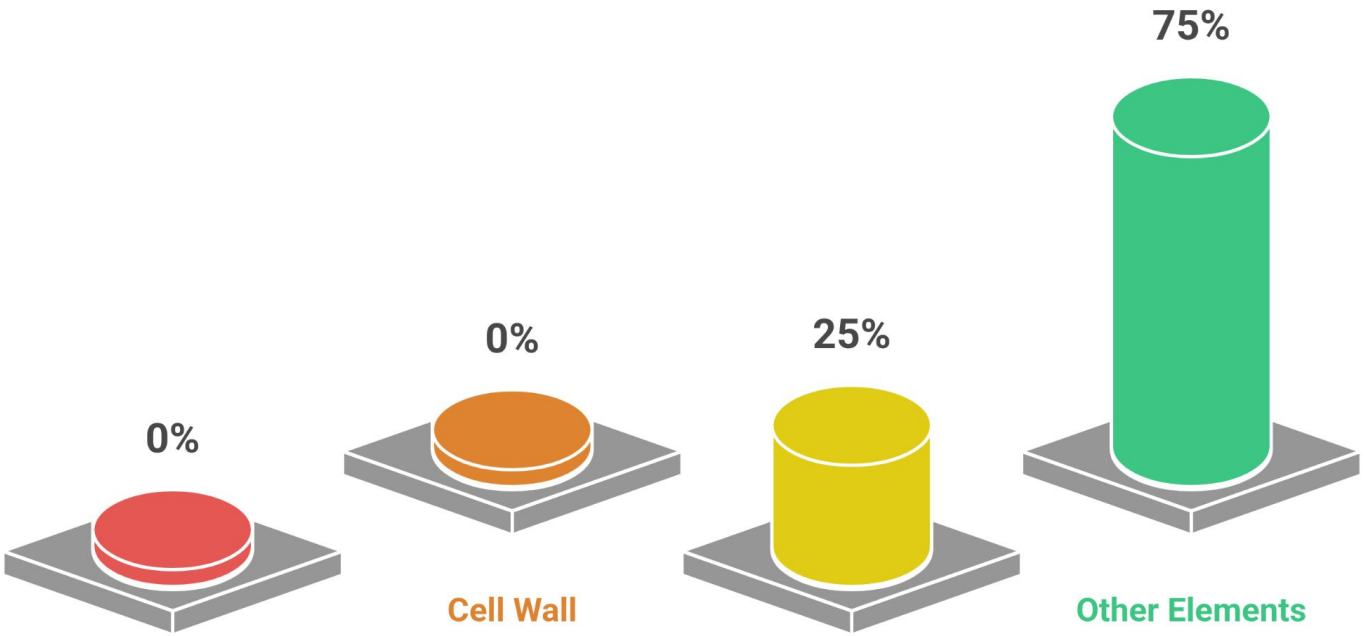
- Subcellular localization helps us understand **where the protein resides inside the bacterial cell**.
- For **drug targets**: cytoplasmic and cytoplasmic membrane proteins are often considered, since they are accessible to small molecules.
- For **vaccine candidates**: extracellular or cell wall proteins are usually prioritized because they are **exposed to the host immune system**.
- Since we are focusing on **drug target identification**, we specifically looked for **cytoplasmic membrane proteins**, as they are essential for bacterial survival and accessible for inhibition.

How we performed it (Process):

1. We used **PSORTb**, a widely used bacterial subcellular localization prediction tool.
2. Input dataset = **60 essential virulent proteins** (from Step 6).
3. PSORTb categorized them into possible locations: extracellular, cell wall, cytoplasmic, cytoplasmic membrane, etc.
4. We selected **cytoplasmic membrane proteins** as the final filter because:
    - They are essential and conserved.
    - They are accessible to therapeutic drugs.
    - They are less prone to antigenic variability compared to extracellular proteins.

Results:

- **Total proteins analyzed:** 60
- **Localization breakdown (PSORTb):**
  - Extracellular → 0 (0%)
  - Cell wall → 0 (0%)
  - Cytoplasmic membrane → 15 (25%) ✅ (selected for next step)
  - Others (remaining 75%) → filtered out
- **Final selected proteins: 15 cytoplasmic membrane proteins**


0%
0%
25%
75%
Extracellular Components
Cell Wall
Cytoplasmic Membrane
Quarter of cellular structure
Other Elements
Majority of cellular structure

Step 8: Enzyme Identification

Enzyme Target Identification

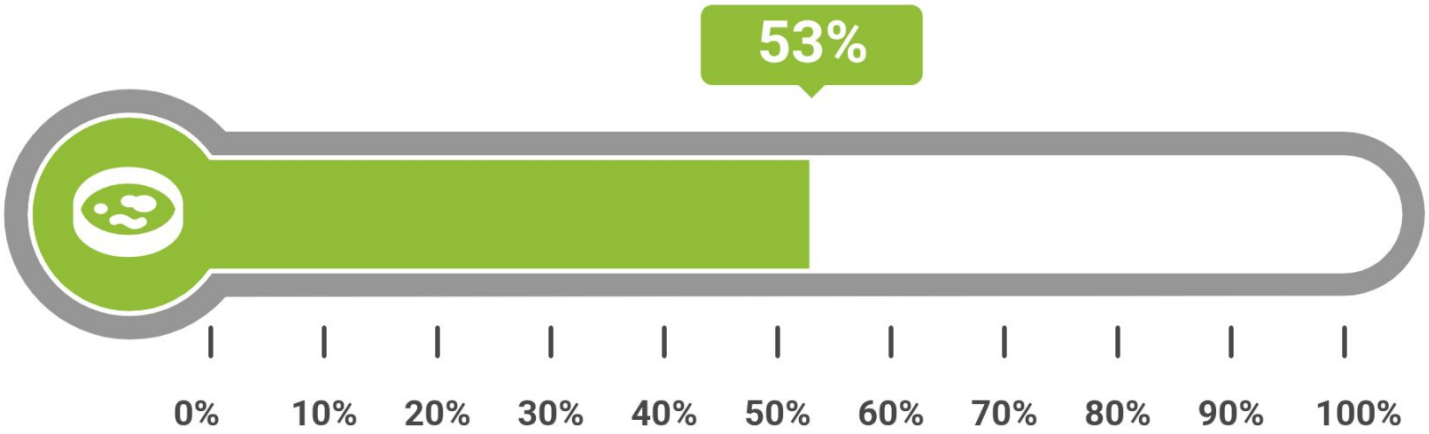
Why we did it (Relevance):

- Enzymes catalyze **critical biochemical reactions** in bacterial metabolism.
- Targeting essential enzymes can **disrupt key metabolic pathways**, making them ideal **drug targets**.
- Not all proteins are enzymes; identifying enzymes ensures we focus on proteins that are **druggable and functionally critical**.

How we performed it (Process):

1. Input dataset = **15 cytoplasmic membrane proteins** (from Step 7).
2. Used **InterProScan**, a tool that integrates multiple protein signature databases (Pfam, SMART, TIGRFAMs, etc.) to **annotate protein families, domains, and enzymatic functions**.
3. InterProScan output provided information about:
    - Enzyme classification (EC number)
    - Protein family
    - Functional domains
4. Filter applied: **Only 8 proteins annotated as enzymes** were retained for downstream analysis.

Protein Analysis



Enzymes Identified

8 out of 15 proteins were enzymes.

Step 9: Hub Protein Identification

Hub Protein Identification

Why we did it (Relevance):

- Proteins rarely act alone; they function as part of **interaction networks**.
- **Hub proteins** are highly connected nodes in a protein-protein interaction (PPI) network.
- Targeting hub proteins can have **maximal impact** on bacterial physiology and survival.
- STRING database provides **high-confidence interaction networks** for known and predicted PPIs.

How we performed it (Process):

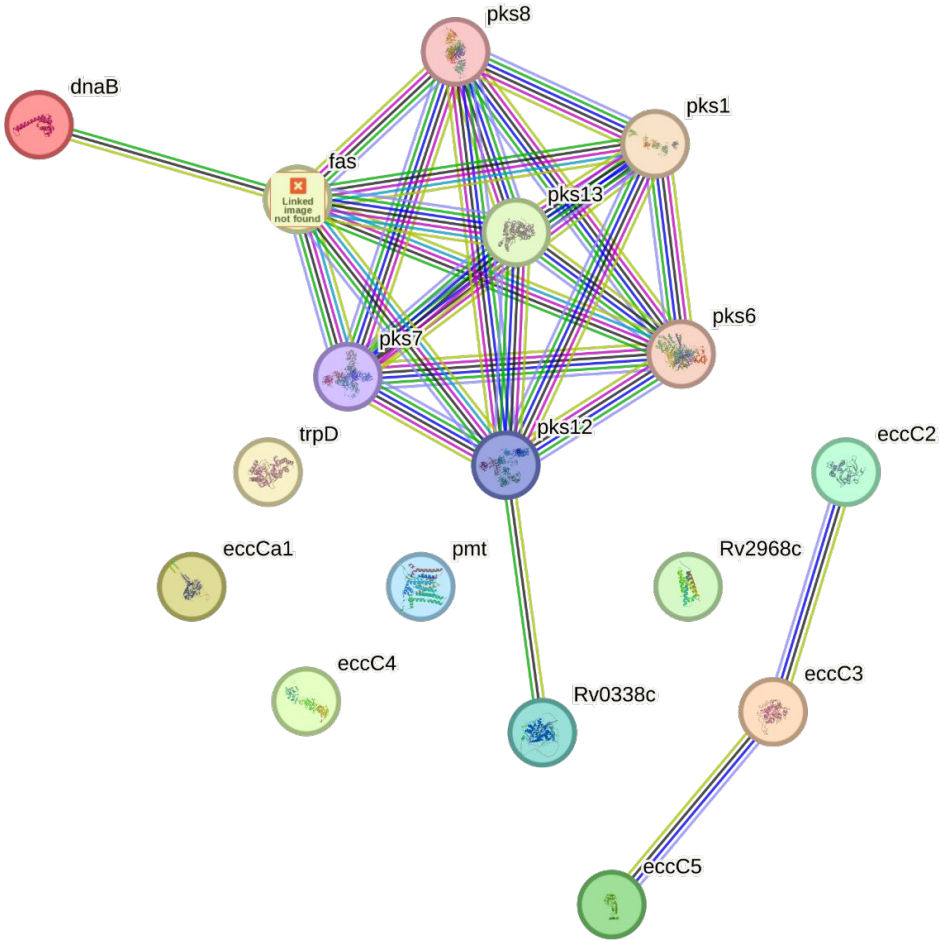
Input dataset = **8 enzyme targets** (from Step 8).

Generated the **STRING network file** (`string_interactions_short.tsv`).

Calculated **degree centrality** for each protein:

- Degree = number of direct interactions with other proteins in the network.

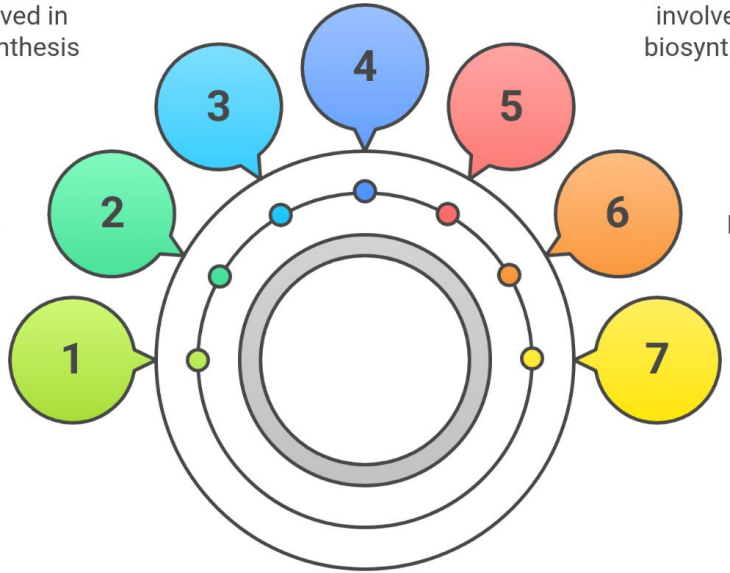
High-degree proteins considered as **hub proteins** for downstream characterization.


pks8
dnaB
pks1
fas
Linked image not found
pks13
pks6
pks7
trpD
pks12
eccC2
eccCa1
pmt
Rv2968c
eccC4
Rv0338c
eccC3
eccC5

Protein Interaction Degrees

Interaction Degree	fas	pks12	pks13	pks1	pks7	pks8	pks6
Degree of Interaction	6	6	6	5	5	5	3

Protein Interaction Network



Protein Sequence Relationships

NP_216178.1

Includes PKS1, PKS7, PKS8, and PKS12



NP_217040.1

Includes PKS13 and FAS

Step 10: Chemical Characterization

Chemical Characterization

Why we did it (Relevance):

To select **druggable proteins**, it is important to consider their **physicochemical properties**.

Parameters such as:

- **Molecular Weight (MW)** – impacts solubility and stability.
- **Instability Index** – predicts protein stability in vitro.
- **GRAVY (Grand Average of Hydropathy)** – indicates hydrophilicity or hydrophobicity.
- **Aliphatic Index** – related to thermostability.

How we performed it (Process):

- Performed **chemical characterization** on the 7 hub proteins belonging to 2 sequences using **BioPython's ProtParam**.
- Filtered proteins based on **stability and physicochemical suitability**.
- Prioritized proteins with favorable **instability index (<40)**, reasonable MW, and balanced hydrophobicity.

Feature	NP_216178.1	NP_217040.1
Sequence Length	1602	3069
Molecular Weight	167,093.81	326,250.51
Instability Index	33.17	32.73
GRAVY	0.26	-0.011
Aliphatic Index	102.12	94.50

Step 11: 3D Structure Prediction

Tertiary Structure Prediction

Rationale:

- Our final two target proteins (NP_216178.1 and NP_217040.1) are **very large** (>1500 residues).
- Traditional secondary structure prediction tools like **PSIPRED** have a sequence length limit (~1500 residues).
- Splitting the sequence for secondary structure prediction could compromise structural context and accuracy.

Approach:

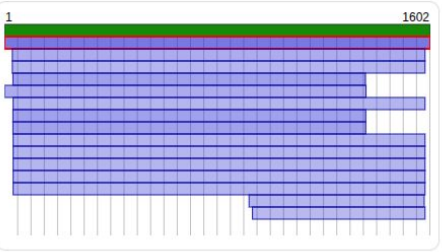
- We **directly submitted the full-length sequences** to **SWISS-MODEL** for **tertiary structure prediction**.
- This allows accurate modeling of the full protein structure and ensures the context of domains and motifs is preserved.
- Secondary structure information can later be **extracted from the predicted 3D model** using tools like **DSSP** if needed.

NP_216178.1 Created: today at 08:09

Summary | Templates 50 | Models 15 | Project Data

Model Results

Order by: GMQE



Model 01



Structure Assessment

Compare

Download files | Display files

Oligo-State
Monomer

GMQE
0.89

Template

O65933.1.A Probable polyketide synthase Pks8

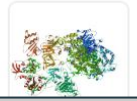
AlphaFold DB model of O65933_MYCTU (gene: pks8, organism: Mycobacterium tuberculosis (strain ATCC 25618 / H37Rv))

Seq Identity
100.00%

Coverage

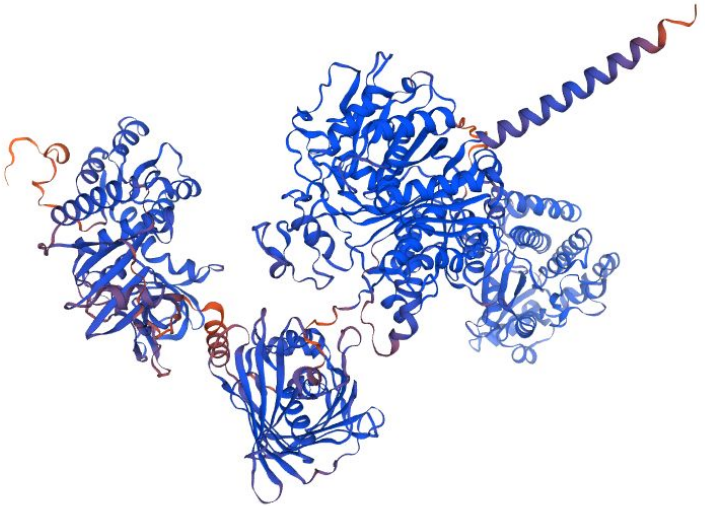
Model-Template Alignment

Model 02



Structure Assessment

Compare



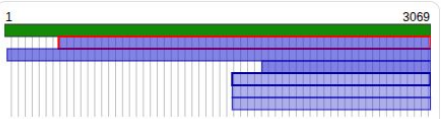
Cartoon

NP_217040.1 Created: today at 08:07

Summary | Templates 50 | Models 6 | Project Data

Model Results

Order by: GMQE



Model 02

Failed to create image for large biounit!

Structure Assessment

Compare ☐

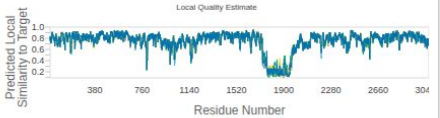
Download files | Display files

Oligo-State Homo-hexamer (matching prediction)

GMQE 0.65

QMEANDisCo Global: 0.75 ± 0.05

QMEANDisCo Local



QMEAN Z-Scores

Template

4v8w.1.A TYPE-I FATTY ACID SYNTHASE

Structure and conformational variability of the Mycobacterium tuberculosis fatty acid synthase multienzyme complex

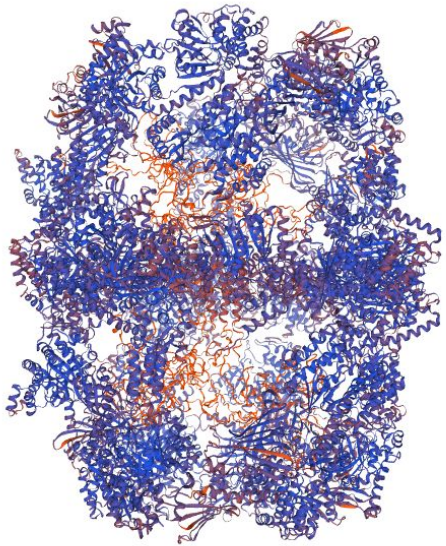
Seq Identity 82.01%

Coverage

Model-Template Alignment

GLY 1166 B

Confidence: 0.77



Cartoon

Step 12: Active Site Prediction for NP_216718.1

Rationale:

- NP_216178.1 was selected for active site prediction due to manageable sequence length.
- NP_217040.1 was **not analyzed** because the protein and file size were too large for the **Protein Plus / DoGSiteScorer** tool.

Tool Used:

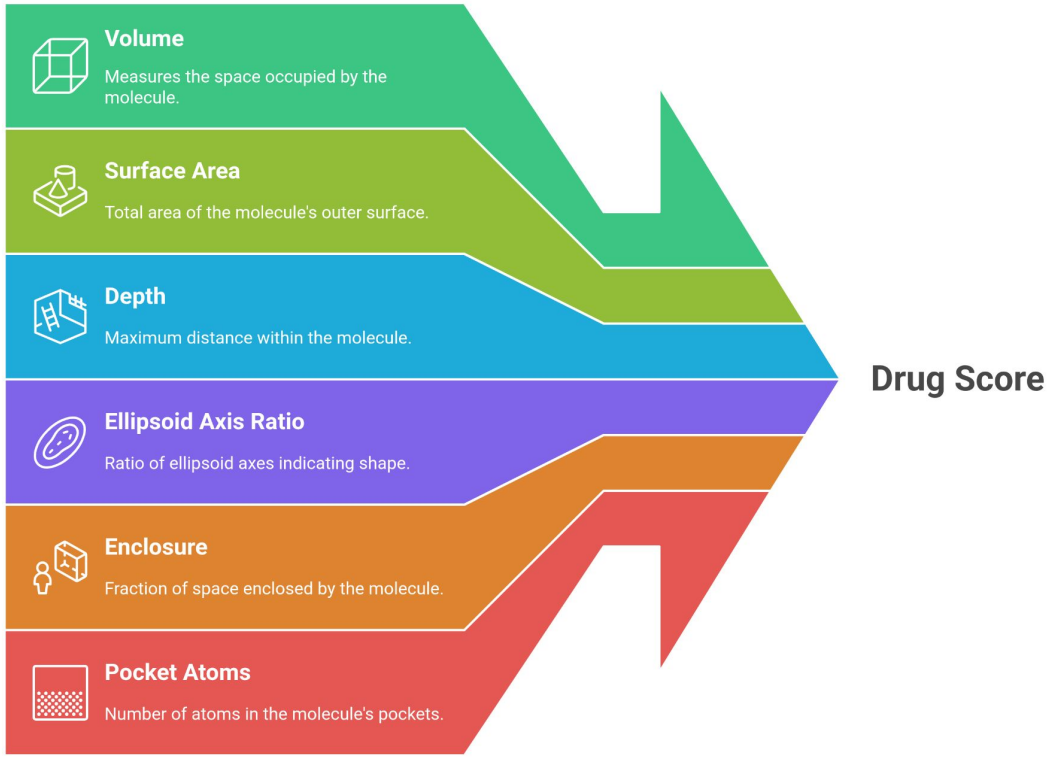
- **Protein Plus (DoGSiteScorer)** – predicts potential binding pockets and evaluates druggability.

Interpretation:

- **High druggability score (0.81)** indicates that P0 is a promising targetable pocket.
- Pocket is **well-enclosed, moderately deep**, and contains **favorable hydrophobic and polar features** suitable for ligand binding.
- Provides a **potential site for small-molecule inhibitor design**.

Next Steps:

- Use this pocket for **in silico docking studies** to screen potential drug candidates.


Volume
Measures the space occupied by the molecule.
Surface Area
Total area of the molecule's outer surface.
Depth
Maximum distance within the molecule.
Ellipsoid Axis Ratio
Ratio of ellipsoid axes indicating shape.
Enclosure
Fraction of space enclosed by the molecule.
Pocket Atoms
Number of atoms in the molecule's pockets.
Drug Score

		Name	Volume Å³	Å²	Score	Score	information
		P_0	1064.9	1267.02	0.81	0.63	Click here to show/hide

Size and shape descriptors

volume [Å³]	1064.90
surface [Å²]	1267.02
depth [Å]	26.84
ellipsoid main axis ratio c/a	0.25
ellipsoid main axis ratio b/a	0.40
enclosure	0.14

Element descriptors

# pocket atoms	246
# carbons (C)	166
# nitrogens (N)	44
# oxygens	35
# sulfurs (S)	1
# other elements	0

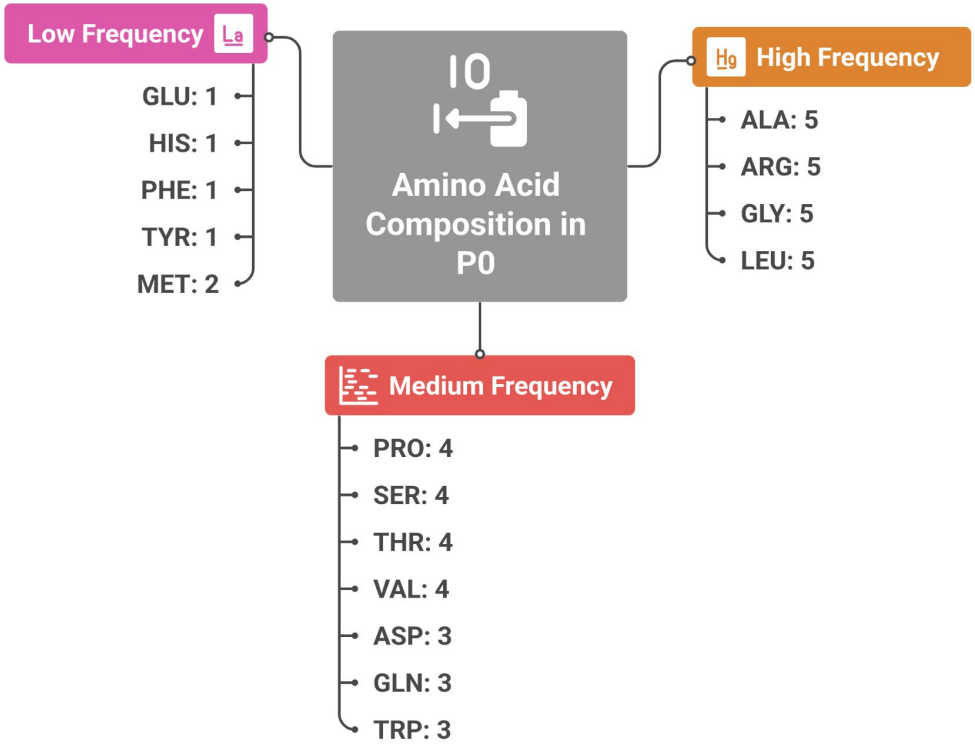
Functional group descriptors

# hydrogen bond donors	30
# hydrogen bond acceptors	63
# metals	0
# hydrophobic interactions	51
hydrophobicity ratio	0.35

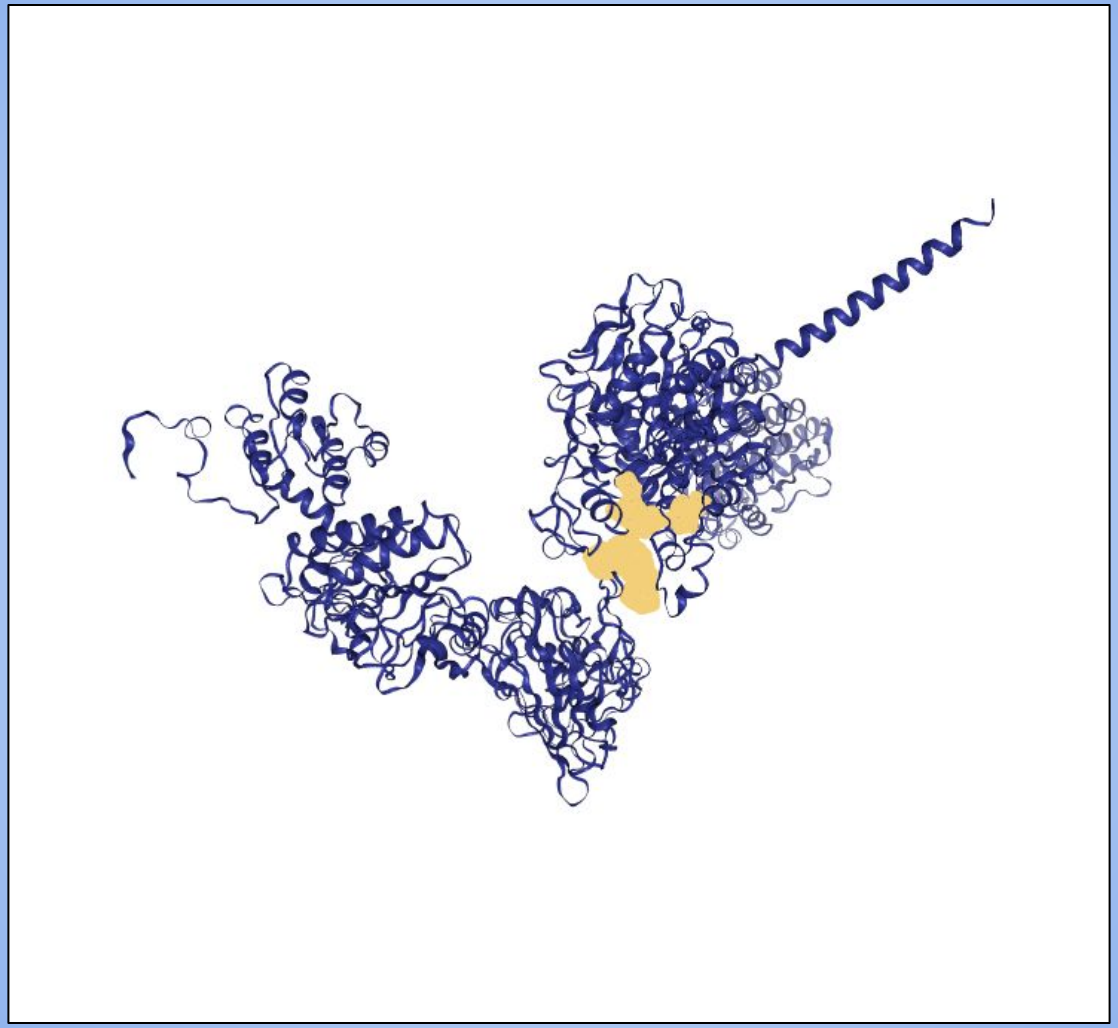
Amino acid composition

apolar amino acid ratio	0.47
polar amino acid ratio	0.33
positive amino acid ratio	0.12
negative amino acid ratio	0.08

Amino Acid Composition in P0



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Result/Conclusion

From an initial **core set (34,765 sequences)** → stepwise filtering produced **2 prioritized targets**:

- **NP_216178.1 (pks cluster)** — preferred (tractable size, druggable pocket)
- **NP_217040.1 (fas)** — very large, keep for later / domain-level targeting

These targets are: **core-confirmed**, **non-homologous to human**, **essential**, **virulent**, **membrane-associated**, **enzyme / hub** proteins.

Advantages of our Pipeline

- Subtractive proteomics reduces off-target risk (host/human filters).
- Uses a high-quality reference to confirm core membership (improves reliability).
- Cross-platform steps (BLAST, FASTA, Python) — accessible to non-Linux users.
- Combination of functional (InterPro), network (STRING), structural (SWISS/DoGSite) evidence increases confidence.

Limitations

- In silico predictions need experimental validation (essential assays, enzyme assays).
- Some web tools limit sequence length (e.g., PSIPRED/SWISS limitations for very large proteins).
- VFDB / pbit / some databases had accessibility issues — required workaround.
- Our core clustering counted per-genome members (34,765) — better ortholog collapsing could change numbers.

Next Steps/Future Work

- Finalize SWISS-Model structures and validate (Ramachandran, QMEAN).
- Dock known inhibitors / small-molecule library into NP_216178.1 pocket → virtual screening.
- MD simulations for top docked ligands.
- Experimental validation: enzyme inhibition + bacterial growth assays.
- Optionally: break NP_217040.1 into domains for modelling and screening.

Thank You

Any Question?